

Title Page

Manuscript title:

VEIN: On-Chain Causal Risk Attribution for Cryptocurrency Systemic Risk

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Author's Information

Quang-Vinh Dang. Quang-Vinh Dang is a Lecturer at British University Vietnam and Data/AI Director at SmartOSC, where he leads initiatives in multi-agent AI systems, large language models, and graph neural networks across banking, retail, and finance. He holds a Ph.D. in Computer Science from Université de Lorraine (LORIA), France (2018), an M.Sc. in Financial Engineering from WorldQuant University (2021), and completed a Post-Master programme in Software Technology at Eindhoven University of Technology. His research spans machine learning in finance, natural language processing, intrusion detection, and federated learning, and he has authored more than 60 peer-reviewed papers indexed in Scopus/SCI-E, many in Q1 journals. Prior to his current roles, he held data-leadership positions at Trustring Social, Be Group, GoTymeX, and DatVietVAC, and was a postdoctoral researcher at Inria, France.

Cover Letter

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The Editor
Financial Innovation

Dear Editor,

I am pleased to submit my manuscript, “**VEIN: On-Chain Causal Risk Attribution for Cryptocurrency Systemic Risk**,” for consideration as a research article in *Financial Innovation*.

The paper proposes VEIN (Verifiable Entity-resolved Interventional/counterfactual Network risk), a framework that treats the resolved on-chain transaction ledger as an auditable candidate structural skeleton for a Pearl-style structural causal model of cryptocurrency systemic risk, rather than inferring the contagion graph statistically from price co-movement as prior causal measures do. On this graph the paper defines an interventional risk measure (On-Chain Causal CoVaR) and, to the best of our knowledge, the first on-chain counterfactual (Pearl Level-3) systemic-risk attribution measure proposed for crypto markets, which decomposes a realised loss into the Shapley-exact share attributable to each upstream entity’s distress. The complete pipeline—a four-tier entity-resolution layer over Dune Analytics’ decoded Ethereum data, the structural causal model, and the risk measures—is implemented and evaluated on real on-chain data around the October 2025 cascading liquidation event, with generality checks on the FTX (2022) and USDC/SVB (2023) stress episodes, and is subjected to a validity and falsification battery (placebo controls, unobserved-confounding robustness at both the coefficient and ranking level, resolution robustness, and external validity against the documented event narrative).

We believe this manuscript is a strong fit for *Financial Innovation*’s scope at the intersection of blockchain, computational financial intelligence, and financial risk management and analysis. The paper combines a methodological contribution (an exact, order-robust counterfactual attribution measure) with an extensive real-data empirical evaluation, and reports negative and suggestive results as candidly as positive ones.

This manuscript is original, has not been published previously, and is not under consideration at any other journal.

Thank you for considering this manuscript. I look forward to your response.

Sincerely,

Quang-Vinh Dang
British University Vietnam

Declarations

Ethical Approval and Consent to participate. Not applicable: this study uses only public, aggregated on-chain and market data and involves no human subjects, human data, or animals.

Consent for publication. Not applicable.

Availability of supporting data. The complete implementation, the real-data evaluation pipeline, and all result artefacts underlying every number, table, and figure in this paper are publicly available at <https://github.com/vinhqdang/VEIN-On-Chain-Causal-Risk-Attribution-for-Cryptocurrency-Systemic-Risk>. The repository includes the full source code, cached query results, and scripts that reproduce every reported result. All underlying blockchain and market data are public and were obtained from Dune Analytics (<https://dune.com>), DefiLlama

(<https://defillama.com>), and Etherscan (<https://etherscan.io>); no restricted or proprietary data were used.

Competing interests. The author declares no competing interests.

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Authors' contributions. Q.-V.D. conceived the study, developed the VEIN framework and its implementation, performed the data collection and empirical analysis, and wrote the manuscript.

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Authors' information (Optional). See “Author’s Information” above.